



Epidemiology of molecular probes in Xpert MTB/RIF assay in Khyber Pakhtunkhwa, Pakistan

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Abstract

Regardless of a plethora of advanced diagnostics, TB and drug resistance remains a principal killer. We proposed gold nanoparticles (AuNPs) attached with probes to enhance the efficiency of GeneXpert MTB/RIF assay instead of conventional dye probes for molecular detection. A total of 15,000 samples were collected from TB suspects and subjected to Xpert MTB/RIF assay, where 6800 (45.3%) were detected as MTB positive, 280 (4.3%) were detected to harbor mutations in the RRDR, while invalid/errors were found in 690 (4.6%) cases. The mutations were detected by probe E, 199 (71.1%), while probes B and D, 30 and 26 (10% and 9%), respectively. In the Xpert MTB/RIF Assay were found mutations picked by probes E and B codons 529–533 (71%) and 512–518 (10%), respectively. The fast-rising works of TB nano-diagnostics, of Xpert probes, may improve by the applications of gold nanoparticle probes.

Keywords Xpert MTB/RIF assay · RRDR · AuNPs · Probe

Introduction

The increasing trend of tuberculosis (TB), worldwide especially multidrug-resistant (MDR-TB), has become a major health problem (Chiang et al. 2013). The causative agent of TB is *Mycobacterium tuberculosis* and the

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simultaneous resistance to two potent drugs, isoniazid (INH), and rifampin (RIF) is termed as MDR-TB. It has been confirmed that relapse patients are more prone to Multidrug-resistant MDR-TB (Ardizzoni et al. 2015). The strain resistant to rifampin has been detailed that *M. tuberculosis* resistant to RIF is more probable to have associated resistance to INH, making RIF resistance a surrogate marker of MDR-TB (Ochang et al. 2016). For proper management of resistant TB, early diagnosis and rapid detection of RIF resistance are important (Brossier et al. 2009). To increase case detection rate, major effort has been done but instead of all these efforts, one-third of new TB cases are still missed because of the lack of rapid, diagnostic facilities in low-income and high-TB-burden countries (Lawn and Nicol 2011). In the treatment of TB rifampicin play a pivotal role among all other drugs (World Health Organization 2013). The patient shows resistance to rifampicin (RR-TB) drug requires long treatment with less compelling and more harmful second-line drugs (WHO 2014). Legitimate treatment of RR-TB depends on provoke conclusions (Organization and others 2016). RR-TB is declared when there is a mutation in the hotspot region (codons 507–533) of the *rpoB* gene and the region called rifampicin resistance-determining region (RRDR) (Zaw et al. 2018). Nucleic acid amplification tests (NAAT) have long held incredible promise for the rapid detection and accurate diagnosis of resistance TB (Pai et al. 2009). In 2008 for the first time, the World Health Organization (WHO) endorsed a nucleic acid amplification test for the use of low-income countries to overcome the global challenge of the resistant TB epidemic. For this purpose after receiving positive signals, an evidence line probe assay (LPA) was endorsed for rapid diagnosis and management of drug-resistant TB (WHO 2008). Another point-of-care rapid molecular diagnostic assay was endorsed by WHO for the simultaneous diagnosis of rifampin resistance within 100 min of the period GeneXpert (Lawn and Nicol 2011). The test requires negligible technicalities and minimal biosafety requirements. This test is a real-time polymerase chain reaction using molecular beacon technology, depends on hybridization, and contains both wild-type and mutant probes. In the Xpert assay mutation occurred in the five overlapping molecular beacon probes (A, B, C, D, and E) covers the 81 bp region of the beta-subunit of the RNA polymerase enzyme *rpoB* gene. (Reddy and Alvarez-Uria 2017). The five overlapping probes can detect mutations in the codons 507–511 (Probe A), 511–518 (Probe B), 518–523 (Probe C), 523–529 (Probe D), and 529–533 (Probe E). Troubleshooting of results displayed in groups according to the manufacturer's instructions: "invalid" (failure of sample processing control because the sample was not properly processed or PCR was inhibited), "error" (failure of the probe check control because the reaction

tube was filled improperly because there was a reagent probe integrity problem, or because the maximum pressure limits were exceeded or there was a GeneXpert module failure) and "no result" (the test was stopped due to power outage) (Reddy and Alvarez-Uria 2017).

Initially, Xpert MTB/RIF assay was endorsed only for the pulmonary (sputum) samples but later on, extrapulmonary specimens were also endorsed for processing (Organization 2013). In the global TB prevalence, Pakistan has been placed fifth among the top 30 high-burden TB countries. According to WHO guidelines, the incidence of Multidrug-resistant TB in new and previously treated cases is 4.2% and 16%, respectively (WHO 2017). In Pakistan, there are very limited data available; so for the strengthening of the *rpoB* gene mutations in Pakistan, Xpert MTB/RIF assay was conducted to diagnose RIF resistance in MTB strains (Ocheretina et al. 2014).

In the current study, we elaborate on the GeneXpert MTB/RIF assay and also describe the molecular epidemiology of RR-TB in Khyber Pakhtunkhwa (KPK) Pakistan. Most terrorism affected province of Pakistan with a total population of 30 million shared the boundary wall with Afghanistan. This is the first detailed report from all Programmatic Management Drug-Resistant TB (PMDT) from this region where the Xpert MTB/RIF method has been used for the detection of over-all RR-TB. The main objective of the study was to identify the mutations, associated with the RRDR comprising of five overlapping probes (A, B, C, D, and E) in the entire MTB genome.

Although, these probes are designed for the identification of precise resistance-associated mutations in the hot spot region are rarely reported in the literature so far, the frequency of these mutations in the specific region could provide useful geographical and epidemiological information about RR-TB. We also suggested a novel method for designing Xpert MTB/RIF probes with gold as nanoparticles for diagnostic purposes. In previous studies, 35 TB nanoparticle (NP)-based assays have been characterized. The most common properties of NP include low-cost, accuracy, and short time result interpretation. Tuberculosis (TB) nano-diagnostics is now applied in the majority of clinical situations, like active stage TB, latent TB, rifampicin-resistant TB, TB/HIV co-infection, extrapulmonary TB, and non-tuberculous mycobacteria (Mirkin et al. 1996; Elghanian et al. 1997; Baptista et al. 2006). The nano-diagnostics is more specific in terms of diagnostic in molecular biology as well as accuracy.

Nanoscience is finding many useful applications in biomedical research particularly in the field of bio-diagnostics. Several assays have been developed with DNA or protein gold nanoparticles for diagnosis (El-Samadony et al. 2017; Singh et al. 2018). This is the first study ever been conducted to provide baseline data applicable to similar studies

in other provinces in Pakistan. Data concerning these mutations would be helpful in the development of novel therapies and more sensitive diagnostic procedures against TB.

Materials and methods

Ethical clearance

Ethical clearance was approved by the Provincial TB Control Programme Research Committee Peshawar, Pakistan (TB/KP/R&D-25-17).

Clinical samples

A total of 15,000 suspected TB samples were collected from January 2017 to December 2017 from four Programmatic Management of Drug-Resistant Tuberculosis (PMDT) units of Khyber Pakhtunkhwa (KPK), PMDT Lead Reading Hospital (LRH), Ayub Teaching Hospital (ATH), SWAT and Mufti Mehmood Memorial Teaching Hospital (MMMTH). Xpert MTB/RIF assay was processed at Provincial Tuberculosis Reference Laboratory (PTRL) accredited as a BSL-III facility. A separate clearance was not required for the current study as these observations are part of a tuberculosis control program.

All the specimens were brought through courier with a well-maintained cool chain box and were further decontaminated using the *N*-acetyl-L-cysteine-sodium citrate-NaOH mechanism (NALC-NaOH) (Kent and Kubica 1985). The sediments obtained after decontamination were suspended in 3 ml, 0.067 M phosphate buffer (pH8.3). These decontaminated specimens were subjected to Ziehl–Neelsen (ZN) staining and carried out microscopy, and also used BACTEC MGIT 960 system for culture and Xpert MTB/RIF assay following the manufacturers' guidelines.

Xpert MTB/RIF assay and molecular beacon technology

The Xpert MTB/RIF assay detects the target sequence of DNA amplified in an automated RT-PCR machine that utilizes molecular beacon technology (Tyagi and Kramer 1996; WHO 2008; Lawn and Nicol 2011). The system comprises five different overlapping probes. These probes labeled with distinct fluorophore dyes target different genomic sequences within the entire hot spot region of the *rpoB* gene (Piatek et al. 1998, 2000). The molecular beacons consist of two arm sequences on both ends of the oligonucleotide probe sequence. These two arms are complementary to each other, to form a loop and stem secondary structure (Tyagi and Kramer 1996). A fluorophore and non-fluorescent quencher are covalently associated with the ends of two arms. Probes

in the non-hybridized state do not emit fluorescence but they bind to a complementary sequence on MTB DNA resulting in conformational changes and cause bright fluorescence emission (Tyagi and Kramer 1996). Any mutations in core *rpoB* sequences affect the hybridization and retain the conformational integrity of the probe resulting in no fluorescence. Using the genomic DNA of *M. tuberculosis* strains, this prototype assay was found to have high specificity and sensitivity for the apprehension of rifampicin resistance (Piatek et al. 2000). The Xpert MTB/RIF assay was run according to the manufacturer's instructions (Cepheid, Sunnyvale, CA) on all processed samples. The probes can detect mutations in the codons 507–511 (Probe A), 511–518 (Probe B), 518–523 (Probe C), 523–529 (Probe D), and 529–533 (Probe E). All generated results were accessed from the attached Xpert (DX) software, including those for MTB-negative or -positive controls. Results were then scored as medium (M), low (L), very low (VL) or high (H), and also whether a sample was rifampicin susceptible or resistant (Boehme et al. 2010).

Drug susceptibility testing

Phenotypic Drug susceptibility testing (DST) was carried out semi-automated BACTEC MGIT 960 system (BD Diagnostic Systems, NJ, and USA) by following the policy statement of the World Health Organization (WHO 2014).

DNA probes with gold nanoparticle

In the recent era, a new approach to use gold nanoparticles for diagnostic purposes is more specific and possessing exclusive optical properties as biosensors and also has been found useful in photothermal therapy drug delivery research. These nanoparticles can appear in a variety of colors and emit light of various wavelengths specifically bright resonance light, falling under the visible region. For small (~30 nm) AuNP, the absorption surface plasmon resonance of the blue–green light portion of the spectrum was ~450 nm, while reflection of red light (~700 nm) yielding a red color due to the unique property; these particles have been widely applied as probes for sensing/imaging of proteins, and nucleic acids targets (Mao et al. 2009; Matsishin et al. 2017). The power of detection has been found eightfold greater with the AuNP-Molecular Beacons (MBs) than with conventional dye-labeled molecular beacon probes. However, there are limitations of high equipment cost and trained persons limited in the nanotechnology field (Shan et al. 2009; Mao et al. 2009). Gold nanoparticle 3D (CID_23985) and stable nuclear magnetic resonance (NMR) structures were retrieved for insilico study, and used default parameters Amber99SB minimized the structure using force field with Based on biocompatibility, biodegradability, and

drug delivery across blood–brain barrier (BBB), gold (Au) (CID_23985) NP were selected as standard Nano-drug for docking with DNA (Probes) by Genetic algorithm from Autodock using custom Atom type parameters for gold and minimized using Chem3D ultra. Docking of both compounds, DNA probes, and nanoparticles was performed and shown that gold can fit on a minor or major groove of DNA selected as standard nanoparticle for docking with probes using Autodock package (Forli et al. 2016). Docking was carried on the structure of DNA (Probes) in the presence of selected potential nano-drug as a ligand for molecular modeling.

Results

Xpert MTB/RIF assay

A total of 15,000 TB suspect samples were subjected to Xpert MTB/RIF assay, of which 6800 (45.3%) were detected as MTB (TB POSITIVE), 280 (4.3%) were detected as Rifampicin Resistant(RR), while invalid/errors were found in 690 (4.6%) cases (Fig. 1).

The assay detected mutations in almost all probes. Among RR, the most infected age group was found as 15–24 (Table 1) followed by 25–30 and 45–60. Quantitatively, the assay interpreted 124 (44.2%) cases as medium followed by 84 (30%) as high. Most samples were pulmonary, 259 (92%), while 21 (7.5%) were extrapulmonary, which have also been detected by Xpert assay (Table 1). More females (58.9%) were infected by RIF-resistant MTB than males. Whereas, age groups 15–24 and 25–34 were more exposed to Rifampicin Drugs (Creswell et al. 2014; Pinyopornpanish

Table 1 RRDR patients socio-demographic characteristics

Age	Number	%	Sample type	Number	%
1–14	18	6.4	Sputum	259	92.5
15–24	79	28.2	Pus	8	2.9
25–34	66	23.6	Gastric lavage	3	1.1
35–44	26	9.3	Bronchial wash-ing	3	1.1
45–60	58	20.7	Tissue	2	0.7
61–95	33	11.8	CSF ^a	2	0.7
			Ascetic fluid	1	0.4
			Lymph node	1	0.4
			Pleural fluid	1	0.4
Sex	Number	%	Total	280	
F	165	58.9			
M	115	41.1			
Total	280				
^a MTB-load	Number	%			
M	123	43.9			
H	84	30			
L	39	13.9			
VL	34	12.1			
Total	280				

^aCSF cerebrospinal Fluid, M Medium, H High, L Low, VL Very Lo

et al. 2015; Gous et al. 2015; Gidado et al. 2018; Sharma et al. 2019).

Probes frequency in RRDR

Among the 280 Rifampicin-Resistant (RR) cases, the most common mutations were located in the region of probe E 199 (71.1%) followed by probes B and D, 30 and 26 (10% and 9%), respectively. However, 12 samples were found as rifampicin resistant by all probes (Table 2).

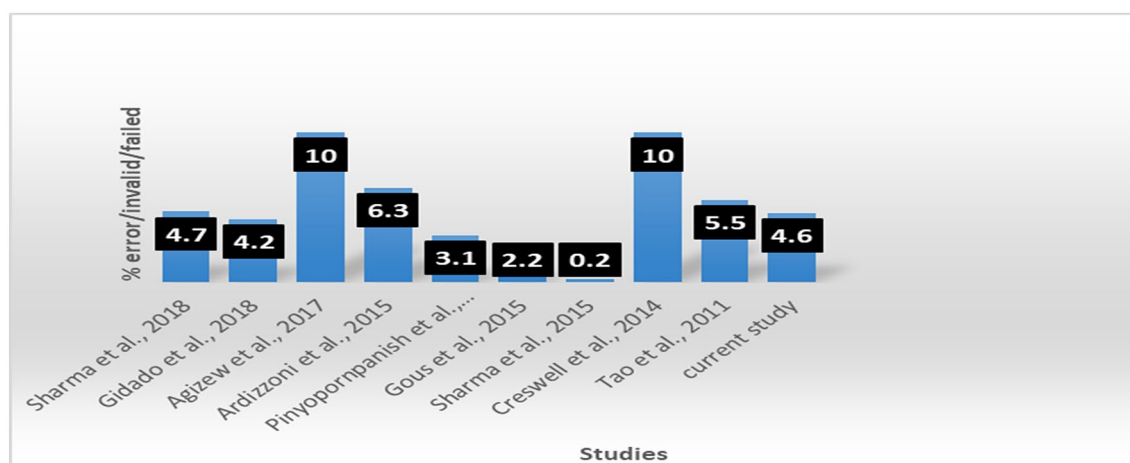


Fig. 1 Error/invalid/failed rate in Xpert MTB/RIF assay detection

Table 2 Frequency of probes among the population of KPK, Pakistan

Probe type	Number	%age
E	199	71.1
B	30	10.7
D	26	9.3
A	6	2.1
C	3	1.1
E&A	2	0.7
B&D	2	0.7
All	12	4.3
Total	280	

Phenotypic drug susceptibility (DST) results were comparatively similar to GeneXpert MTB/RIF assay. Among 280 RR cases, first-line phenotypic DST of RIF, INH, PZA, and EMB was detected as 279 (99.64%), 226 (80.7%), 71 (25.3%), and 52 (18.5%), respectively. These results showed that the RR cases have a significant number of other important first-line INH and PZA resistance (Table 3). Among the total RR, 226/280 (80.75%) were declared multidrug resistance (MDR) on Phenotypic DST, while 67 (2.3%) were resistant to three out of four first-line drugs, RIF, INH, and PZA in combination.

Gold nanoparticles (AuNPs)

To reduce the errors and failure rate in the Xpert MTB/RIF assay, gold nanoparticles (AuNPs) are now gaining importance in TB diagnostics. Based on the simple procedure to synthesize, colorimetric detection and modification, gold nanoparticles (AuNPs) got significant attention from clinicians and researchers. To observe the binding affinity of AuNPs with DNA, we employed docking approaches. A

total of twenty confirmations were observed in docking complex, among which only one was analyzed, fitting best with DNA. We employed a genetic-based algorithm for the NPs to conduct docking with DNA as the selected NPs. Following, only one confirmation was selected from 20 conformations per molecule to analyze the best fitting conformations at the active site of DNA. The docking results revealed that only gold NP showed the best fitting in the active site of DNA (Fig. 2). Gold nanoparticles docking showed a potential binding affinity with -1.1 kcal/mol binding energy. DNA probes labeled with AuNPs showed non-classical hydrogen bonds and TB nano-diagnostics showed promising results to be ideal detection tools that can meet the rigorous demands to end the TB epidemic by 2030 (Fig. 2).

Discussion

The fast-molecular diagnostic techniques are in progress specifically for tuberculosis; however, in the current era, Xpert MTB/RIF assay is an advanced rapid molecular method of detection that is more sensitive and specific (Marlowe et al. 2011).

The GeneXpert is currently detecting 81-bp regions in the *rpoB* gene using molecular beacons. This region is divided into five overlapping parts called the rifampicin-resistant determining region (RRDR). These five probes can detect mutations in the codons 507–511 (Probe A), 511–518 (Probe B), 518–523 (Probe C), 523–529 (Probe D), and 529–533 (Probe E), can be used to detect about 93% of resistance-conferring mutation to rifampicin (Ochang et al. 2016). Unsuccessful results were classified according to the manufacturer's instructions: "invalid" error and No Result (Reddy and Alvarez-Uria 2017). Phenotypic determination

Table 3 RRDR probes and first-line drug resistance

RIF	A	B	C	D	E	B&D	E&A	All	Total
R	6	30	3	26	198	2	2	12	279
S	0	0	0	0	01	0	0	0	1
INH	A	B	C	D	E	B&D	E&A	All	Total
R	5	25	3	19	162	1	2	9	226
S	1	5	0	7	37	1	0	3	54
PZA	A	B	C	D	E	B&D	E&A	All	Total
R	0	9	1	5	54	0	0	2	71
S	6	21	2	21	145	2	2	10	209
ETM	A	B	C	D	E	B&D	E&A	All	Total
R	0	6	0	7	35	0	0	4	52
S	6	24	3	19	164	2	2	8	228
Total	6	30	3	26	199	2	2	12	280

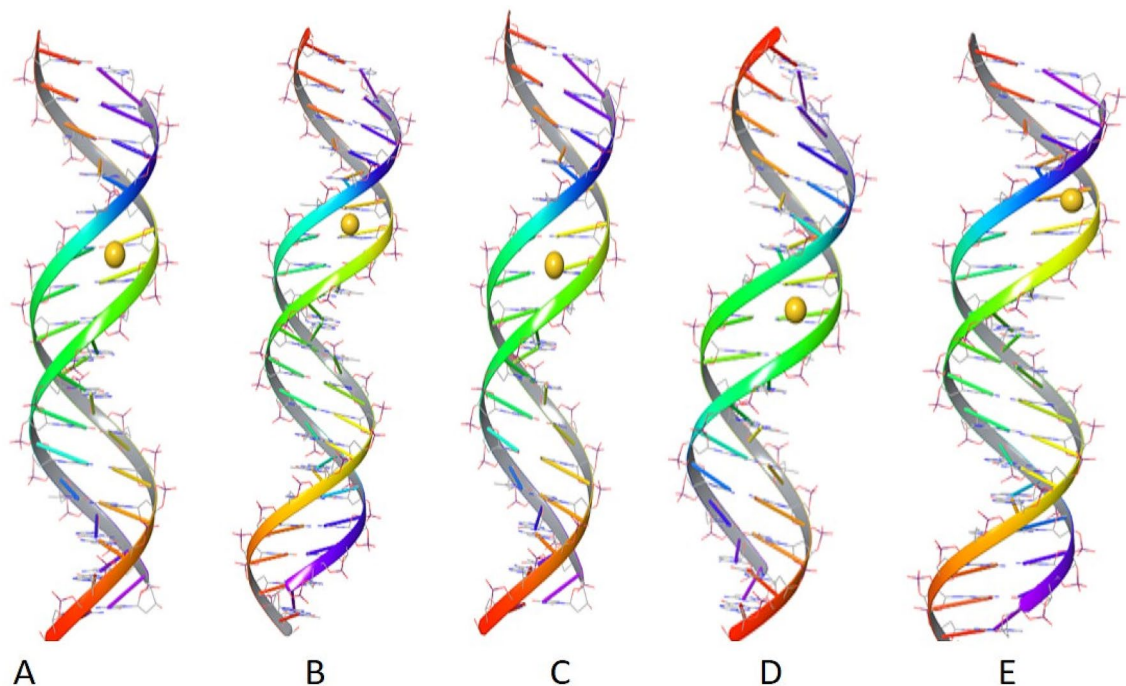


Fig. 2 Representation of Probe (DNA) interaction with gold nanoparticle (Yellow). **a–e** is the types of probes

of resistance to RIF revealed similar results as the Xpert assay, i.e., 279/280 (99.64%) (Table 3). In the current study, the percentage of RR cases is more in the age group 15–24 (28.2%) and 25–34 (23.6%) which is in close agreement with a previous study (Coovadia et al. 2013). In gender-wise comparison, the females (58.9%) were more prevalent in RR than males, maybe due to the cultural restriction of female or geographical variations but still, no scientific evidence was found. Multidrug-resistant (MDR) cases detected by conventional DST among the GeneXpert RR were 205/280 (73.2%) which were quite high. However, 2.3% ($n = 67$) were resistant to three first-line drugs, RIF, INH, and PZA in combination. In a previous study (Huang et al. 2018), the MDR-TB in Xpert MTB RIF resistance cases was detected as 122 (82.99%) by traditional culture and DST techniques.

The mutation was detected in probe E, 199 (71.1%) isolates are the most prevalent in 81-bp region of *rpoB* gene followed by probes B and D as 30 and 26 (10% and 9%). There were 12 cases in which all the probes were detected, while 9 RR samples were detected by more than one probes (Table 2). In a previous study (Yue et al. 2003), the probe regions associated with RR codons are 531 (41%), 526 (40%), and 513 (4%). Therefore, the above study correlates somehow with our findings, that a probe E region is more likely exposed to mutations.

We detected invalid/errors in Xpert MTB/RIF assay in 690 (4.5%) cases; while in a more recent study by Gidado

et al. (2018), 4.2% had error results, 4.7% of the total number of results were invalid, and 2.1% no result outcomes that reflect attention for more sensitive and specific probes could improve the performance of Xpert assay. AuNPs are good candidates in the field of TB diagnostics given their good biocompatibility and unique optical properties such as surface Plasmon resonance (SPR) absorption and resonance light scattering. Furthermore, the relative ease at which their surfaces can be modified with nucleotides enables these AuNPs to function effectively as probes. AuNPs have been successfully modified in the bench setting to more efficiently and rapidly detect the presence of MTB proteins (El-Samadony et al. 2017). We found DNA gold nanoparticles (AuNPs) docking results showed a potential binding affinity with -1.1 kcal/mol binding energy and non-classical hydrogen bonding (Fig. 2). Procedures for NP synthesis and functionalization should be streamlined toward reproducible large-scale production that may be useful to reduce the errors/invalid results. Also, there is a need to optimize protocols for sample collection, transport, and processing especially for DNA extraction to omit the need for target amplification before detection. Technical optimization is also needed to develop fully integrated diagnostic systems to eliminate contamination risk. The close collaboration of multidisciplinary scientists and clinicians, alongside with active involvement of funding agencies and policymakers, can accelerate bench-to-bed-side translation.

Conclusion

In conclusion, the Xpert assay was able to investigate and detect resistance-associated mutation in the RRDR 81-bp region. The most common mutations in the Pashtun dominant region of KPK, Pakistan were found in Probes E and B codons (529–533) (71%), (512–518) (10%), respectively. Further studies are needed to find the resistance correlation of RRDR with phenotypic RIF drug susceptibility testing (DST). Progress in TB nano-diagnostics and the detection capability of Xpert probes may be improved significantly by the application of gold nanoparticle probes. Geographical-specific biomarkers may further assist in drug resistance TB management.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no competing interests.

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